14.3 Local linearity and differentials

A function $z = f(x,y)$ is **differentiable** at (a,b) if when we zoom in on the point $(a,b,f(a,b))$ on the graph we see a "plane." The plane that best matches the surface at $(a,b,f(a,b))$ is called the **tangent plane** at (a,b). Its equation is

$$T(x,y) = f_x(a,b)(x-a) + f_y(a,b)(y-b) + f(a,b).$$

This is also called the **local linearization** of f near (a,b).

→ Do problems 1–4.

The **differential** of $z = f(x,y)$ is

$$\partial z = f_x(x,y)\,\partial x + f_y(x,y)\,\partial y$$

Think of ∂x and ∂y as being small changes in x and y, and then ∂z represents a small change in z. I.e.,

$$\partial z = f_x(a,b)(x-a) + f_y(a,b)(y-b)$$

approximates $f(x,y) - f(a,b)$ if (x,y) is close to (a,b).

→ Do problem 5.